RackBotGen3 (Corvin) system wiring

Sheet 2: robot harness (battery, fuse, switch, INA226, E-stop relay, bucks, charger, Pi, motion board connectors, motors, switches, servos, LED, buzzer).
Sheet 3: rack and spare station (mains, 24 V PSU, rack controller terminals, bay switches, E-stop, dock relay, lamps, station cables, dock contacts and station switches).
Sheet 4: ESP32 rack controller, the PLC fallback (P40), wired to the same terminal names.

The two sides meet only at the spare-station dock: robot pads J102 ↔ station contacts J202 (24 V charging, switched by the rack's dock relay), and over Wi-Fi (Pi AP RACKBOT ↔ rack controller).

Robot harness

File: robot.kicad_sch

Rack and spare station

File: rack.kicad_sch

ESP32 rack controller (PLC fallback)

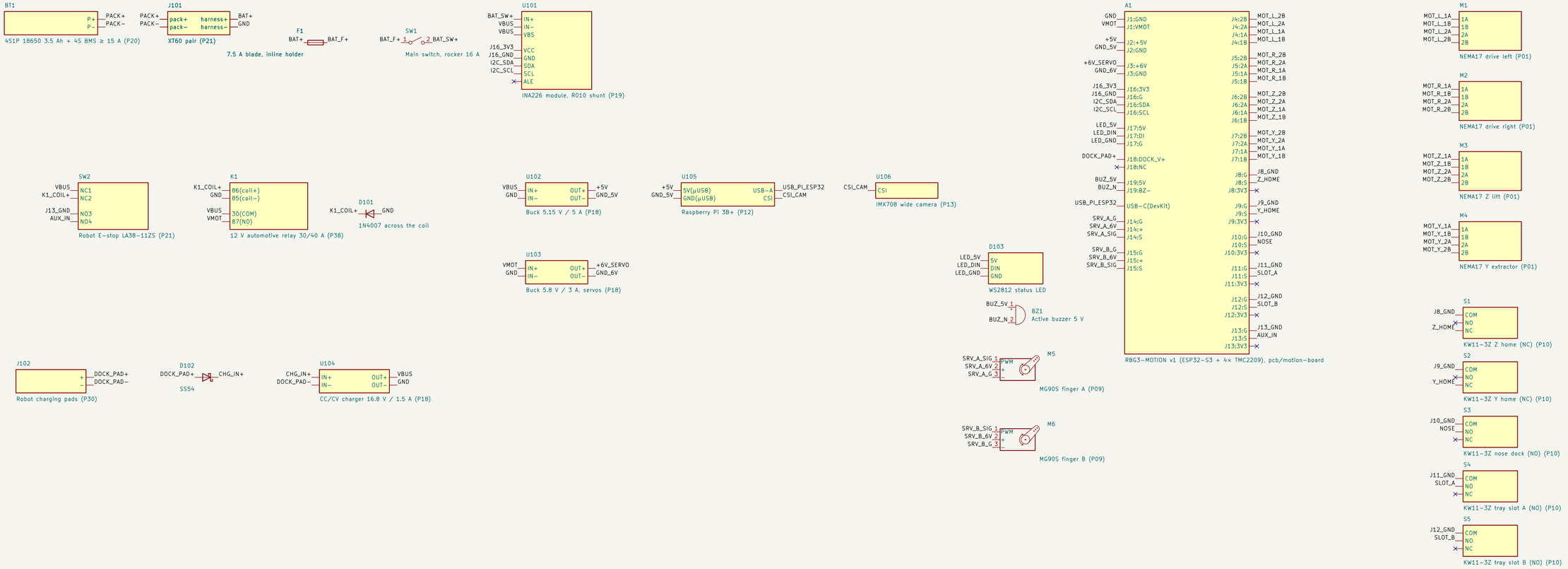
File: rack-esp32.kicad_sch

Board-level circuit: pcb/motion-board/rbg3-motion.kicad_sch
Generated by pcb/system/scripts/gen_system.py; edit the script, not these files
RackBotGen3 (Delta Cup 2026)

Sheet: /
File: rbg3-system.kicad_sch

Title: RBG3 system wiring: overview

Size: A4	Date: 2026-10-24	**Rev: 1**
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Robot harness. Power path: pack → XT60 → F1 7.5 A → SW1 → INA226 shunt → VBUS (12-16.8 V).
VBUS → E−stop NC → K1 coil (K1 makes VMOT) → VMOT → board J1 (XT30) and the servo buck.
VBUS → 5 V buck → PI (micro−USB pigtail) and board J2. The ESP32 is powered over USB from the Pi.
Charging: dock pads → SS54 → CC/CV charger → VBUS (charges the pack through the INA226, current reads −).
DOCK_PAD− goes ONLY to the charger IN−, never to GND: the charger senses current between IN− and OUT−.

Wires (02 §1): VBUS/VMOT and pack 18 AWG silicone red/black (tag VMOT); dock pads → charger 18 AWG;
5 V and 6 V buck outputs 20 AWG; motors 22-24 AWG JST−XH 4p; switches, E−stop NO, DOCK_V 26 AWG JST−XH.
Connectors: XT60 at the pack, XT30 pigtail to J1, 5.08 screw terminals J2/J3, JST−XH J4-J13 and J16-J19.
Ground rule (02 §3.3): GND_5V and GND_6V run from each buck's OUT− to J2/J3, not back to the pack;
keep the J1 GND lead short and thick. GND, GND_5V and GND_6V are one node (inside the bucks and on A1).
J8-J13: switch COM → G, contact → S; the 3V3 pin is spare. Home switches use NC (open = home or broken wire).
Polarity: wire red = +, black = −; meter every lead before the first plug−in (A1 README bring−up).

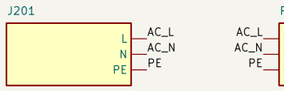
Nets connect by name (local labels): check block pin names on the bought modules
Spec: docs/spec/02−electrical.md §1 (power), §3.2 (board connectors); parts: docs/parts.csv

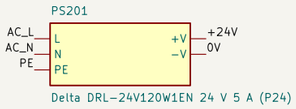
RackBotGen3 (Delta Cup 2026)

Sheet: /Robot harness/
File: robot.kicad_sch

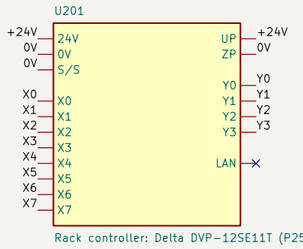
Title: RBG3 system wiring: robot harness

Size: A2	Date: 2026−10−24	**Rev: 1**
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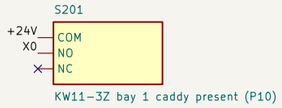


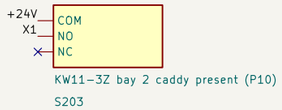


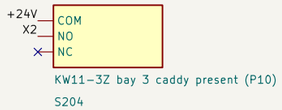
PE: also bond the aluminium rack frame to PE.

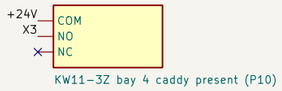


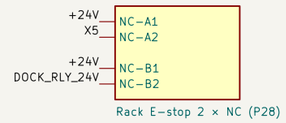
LAN: only with the PLC (Wi−Fi client bridge to the Pi AP RACKBOT, 02 §4.3). The ESP32 uses Wi−Fi.

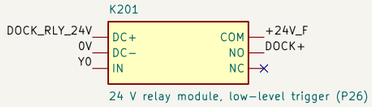




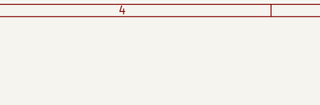


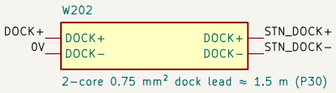






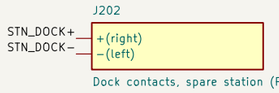












Rack DIN rail: PSU, U201, K201, F201, terminal blocks for +24V / 0V. Spare station: S205-S207 and J202.
Inputs: switches connect +24V to Xn, S/S to 0V (sourcing). Outputs Y0-Y3 sink (NPN): lamps and K201 IN go between +24V and Yn. UP = +24V, ZP = 0V power the PLC's transistor outputs.
Dock power: K201 closes only while Y0 = M0 AND X4 AND X5 AND watchdog OK (ladder, 03 §4). The E−stop's second NC contact feeds K201's DC+, so a pressed E−stop drops the relay in hardware.
J202 mates with the robot's charging pads (robot sheet J102): + on the right, − on the left, viewed from the robot. 24 V reaches the pads only while the robot is docked and X4 is closed.
The rack's own dock has no switch and no contacts (01 §4): the robot's NOSE switch confirms docking there.

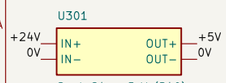
U201 terminals are the PLC's; the ESP32 fallback (sheet 4) exposes the same functions
Spec: docs/spec/02−electrical.md §4 (rack), §5 (safety); 03 §4 (Modbus); 01 §5 (stations)

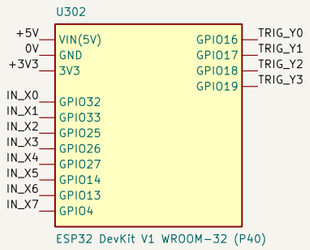
RackBotGen3 (Delta Cup 2026)

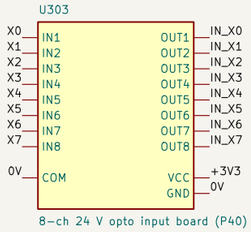
Sheet: /Rack and spare station/
File: rack.kicad_sch

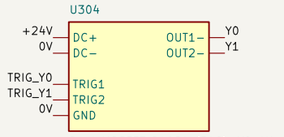
Title: RBG3 system wiring: rack and spare station

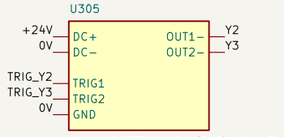
Size: A3	Date: 2026−10−24	**Rev: 1**
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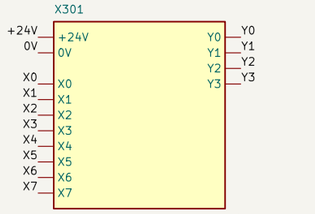












Replaces U201 on the rack sheet: the wires that went to the PLC's X0-X7, Y0-Y3, 24V and 0V land on X301.
Inputs: opto IN1-IN8 take the +24 V switch signals with COM to 0V; the opto outputs pull the GPIOs low when an input is on (use INPUT_PULLUP if the board has no pull-ups; check its jumper/polarity).
Outputs: the D4184 modules sink like the PLC's NPN outputs, so lamps and the dock relay module (low-level trigger) keep their +24V common wiring. GPIO16-19 drive the triggers (3.3 V logic).
Proposed pins avoid the ESP32 strapping pins 0, 2, 5, 12, 15 and the flash pins 6-11; GPIO34-39 are left free (input-only, no pull-ups). The firmware exposes the same Modbus TCP map as the PLC (03 §4) over Wi-Fi.
Module pin names vary by vendor: check the bought boards before wiring.

GPIO choice is a proposal: confirm it when the rack firmware is written
Spec: docs/spec/02-electrical.md §4.4; parts: P40

RackBotGen3 (Delta Cup 2026)

Sheet: /ESP32 rack controller (PLC fallback)/
File: rack-esp32.kicad_sch

Title: RBG3 system wiring: ESP32 rack controller (PLC fallback)

Size: A3	Date: 2026-10-24	**Rev: 1**
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